ALICE L WALTON SCHOOL OF MEDICINE

Thrombotic Disorders: Foundations and Frameworks

Last updated October 27, 2025 16 min read ScholarRx

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Adapted from: Thrombotic Disorders: Foundations and Frameworks

Listen to this Brick

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Learning Objectives (4) Versions

After completing this brick, you will be able to:

1. Differentiate the clinical presentation of thrombotic disorders.
2. List the types of thrombotic disorders and describe their basic underlying causes.
3. Justify the therapeutic roles of warfarin, heparin, and direct thrombin inhibitors in treating thrombotic disorders.
4. Recognize adverse drug effects, drug interactions, and contraindications of warfarin, heparin, and direct thrombin inhibitors.

CASE CONNECTION

LK is a 56-year-old female following up on orthopedic service. She had a right total knee replacement 2 days ago. Prolonged postoperative nausea and vomiting have delayed the start of her physical therapy. She also describes right calf pain and swelling. On examination, her right calf is tender; it is 5 cm larger than the left is when measured 5 cm below the inferior border of the patella. A right-lower-extremity scan with Doppler is ordered, which documents a large deep vein thrombus (DVT). You explain the test results to LK. "I've never had a blood clot before. How did this happen?" she says. "Can it be treated?"

How will you answer LK's questions? What workup should be done? Consider your answers as you read, and we'll revisit LK at the end of the brick.

GO TO CONCLUSION ↓

What Are Thrombotic Disorders (LO #1)?

Thrombotic events (abnormal blood clots) are very common problems. In fact, the most common cause of death in the United States, myocardial infarction (heart attack), is usually caused by a thrombotic event. Many risk factors, such as smoking, cancer, a likelihood to form thrombi.

Sometimes, however, patients with none of these risk factors develop thrombi. They may have an underlying **thrombotic disorder**, which can be hereditary or acquired as a part of an underlying disease such as systemic lupus erythematosus. Although thrombotic disorders are uncommon, it is important to be aware of their mechanisms and clinical features, because once a diagnosis is made, the patient can often be treated effectively to prevent clot formation.

The overall process of thrombosis and its consequences was discussed in a previous brick, Thrombosis and its Consequences.

How Do Patients With Thrombotic Disorders Present (LO #1)?

Most patients who develop thrombi do not have a predisposing thrombotic disorder. However, certain signs in a patient's history and physical exam should raise suspicion of an underlying thrombotic disorder:

- Personal history of recurrent thromboses
- Family history of thrombotic episodes
- Age <50 years
- Personal history of miscarriage
- Presence of a venous thrombus somewhere other than the legs or lungs

Which thrombus localizations in a patient can clue you in to a possible underlying thrombotic disorder?

What Are the Types of Thrombotic Disorders (LO #2)?

Figure 1 shows the mechanisms of several genetic thrombotic disorders as well as one that is acquired: antiphospholipid antibody syndrome.

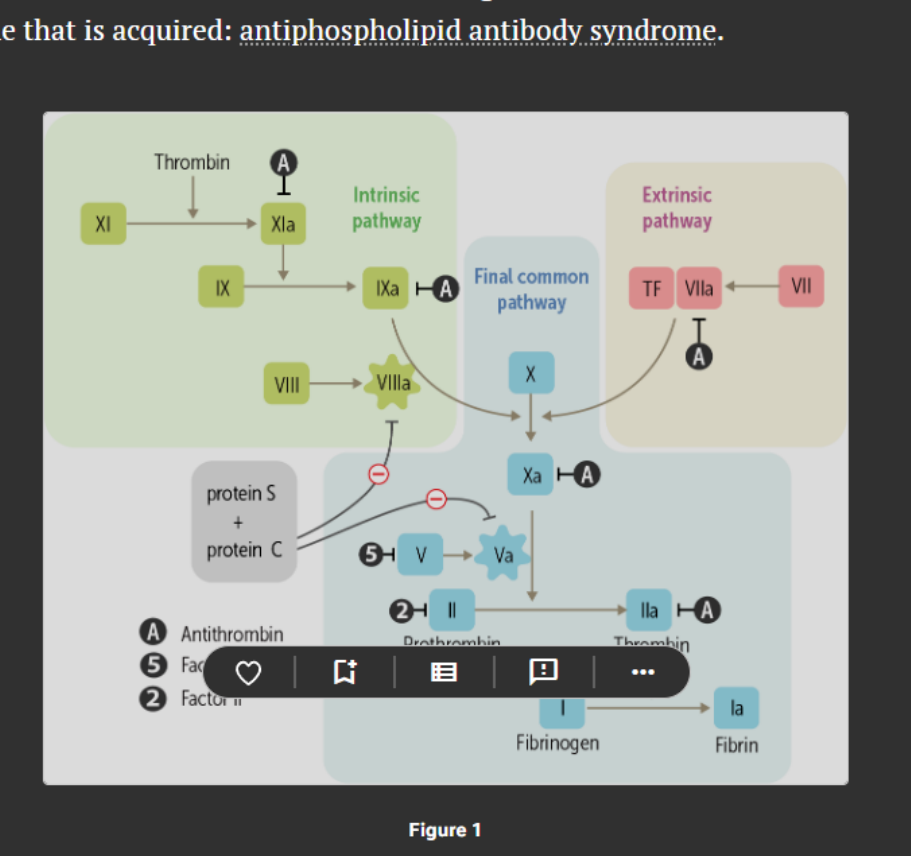

Figure 1

Genetic thrombotic disorders involve one or more pathways of the clotting cascade, leading to either an overactive enzymatic reaction that causes clotting or a decrease in clotting inhibitors.

Please refer to the previous overview of hemostasis and the clotting cascade if you would like further review (Hemostasis).

Genetic Disorders That Stimulate Clotting

We'll cover the disorders resulting from overactive clotting first. Refer to Figure 1 as we go.

Factor V Leiden. This is the most common of all the thrombotic disorders. It's caused by a point mutation in the factor V gene that makes the resulting factor V (factor V Leiden) resistant to inactivation by protein C (see 5 in Figure 1). This means that factor V Leiden participates in the coagulation cascade, but it can't be turned off, and the cascade continues to make fibrin when it shouldn't.

Factor II (Prothrombin) Gene Mutation. In this rare disorder, patients make too much of one of the coagulation factors. Factor II (prothrombin) is a precursor to the important clotting component thrombin (factor IIa), which causes the formation of fibrin and, ultimately, clot formation (see 2 in Figure 1). Too much factor II leads to too much thrombin, which leads to the formation of excess f .

Genetic Disorders That Inhibit Clotting Termination

Now we'll look at thrombotic disorders caused by a decrease in clotting inhibitors.

Antithrombin Deficiency. This is a rare genetic defect in which, as the name suggests, patients make too little antithrombin. Antithrombin is an inhibitor of the coagulation cascade; it acts on most of the factors and shuts down the cascade very effectively (see A in Figure 1).

Protein C and Protein S Deficiencies. These genetic abnormalities disrupt the function of proteins C and S, which act together to shut down the coagulation cascade. The genetic defects, however, are separate diseases. The problem here is the same as in antithrombin deficiency: if you lack a major inhibitor of the coagulation cascade, it's hard to turn the cascade off, and the result is too much fibrin formation.

Acquired Disorders

Antiphospholipid antibody syndrome is the main acquired thrombotic disorder. Patients make antibodies to phospholipids, which in some patients can wreak havoc on the vascular endothelium and on platelets in complex ways, leading to abnormal and excessive clotting. It can happen as a primary disorder without underlying systemic disease, or it can be part of a systemic disease such as systemic lupus erythematosus.

Anticoagulant Therapy (LO#3–4)

Given that genetic an cally present with thromboses, similar treatments are used to treat these conditions: anticoagulant therapy. Two common medications used in these scenarios are warfarin and heparin. The main principles of these two therapies will be discussed below.

Warfarin

Mechanism of Action

Warfarin is classified as a vitamin K antagonist. This means it inhibits hepatic synthesis of the vitamin K–dependent clotting factors. Specifically, warfarin interferes with the γ-carboxylation of vitamin K–dependent clotting factors by inhibiting vitamin K epoxide reductase. The specific vitamin K-dependent clotting factors include *proteins C and S* and *factors II, VII, IX, and X* (Figure 2). Given the effects of warfarin on many different clotting factors, it gives us many considerations on the adverse effects, monitoring, and contraindications that will be discussed below.

Recall that proteins C and S inactivate factors Va and VIIIa, which are regulatory factors. Proteins C and S have a shorter half-life than that of the clotting factors.

This means that initially warfarin, i.e. within the first 24-48 hours, puts patients in a hypercoagulable state. Often, this requires bridging with heparin (see mechanism and more info on heparin below) until the full anticoagulant effects on factors II and X take place at about 72 hours after initiation. The full anticoagulant effect can take between 5-7 days after starting warfarin. This 'delayed' onset is due to the long half-life of warfarin.

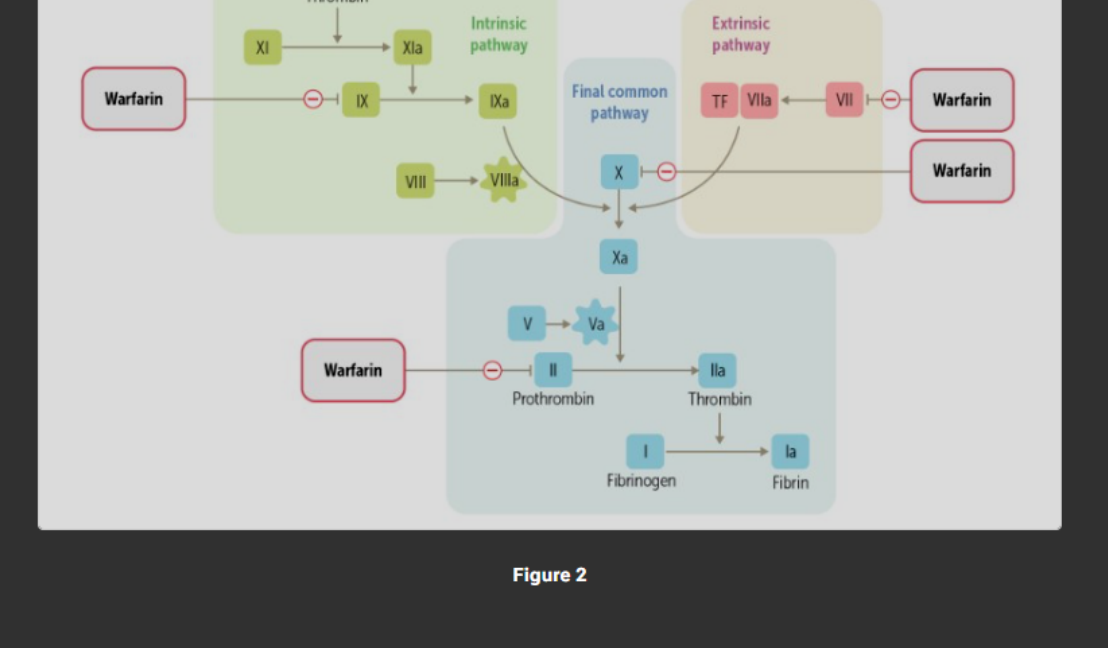

Figure 2

What are the vitamin K-dependent clotting factors?

Drug Monitoring

Due to the risks of warfarin therapy (see below), close monitoring of dosing is needed. A coagulation test, rather than a warfarin drug level, is used because we are principally concerned with how well the drug inhibits clotting. Because warfarin reduces the production of clotting factors in both the extrinsic and intrinsic pathways, the drug will prolong both prothrombin time (PT; mainly monitors the extrinsic and common pathways) and the partial thromboplastin time (PTT; primarily monitors the intrinsic and common pathways).

That said, **warfarin is mainly monitored using the PT** or its clinical equivalent, the **international normalized ratio (INR).** This is because factor VII (part of the extrinsic pathway) has the shortest half-life of all the vitamin K-dependent factors. Given that PT monitors the extrinsic clotting pathway, warfarin prolongs the PT/INR before the PTT, so the PT or INR is an earlier indication of warfarin effect.

The higher the warfarin concentration, the more prolonged is the PT/INR. This becomes important in monitoring bleeding risk.

CLINICAL CORRELATION

In most patients, the desired therapeutic range for INR is 2.0-3.0, like those with thrombotic disorders. In patients with mechanical prosthetic heart valves, the desired therapeutic range is a little higher (2.5-3.5).

Adverse Effects

The most concerning adverse effects of warfarin are excessive bleeding and skin necrosis.

Excessive Bleeding. Warfarin has the potential to cause *severe bleeding.* There are two ways to reverse excessive bleeding caused by warfarin. Often, these two therapies are used simultaneously.

- **Prothrombin Complex Concentrates (PCC)** - This medication essentially replaces the clotting factors that warfarin depletes (factors II, VII, IX, and X, as well as Proteins C and S) to reverse the cause of bleeding.
- **Vitamin K** - As discussed above, warfarin inhibits the synthesis of vitamin K–dependent factors. Giving Vitamin K will not reverse severe bleeding, however, it will replete Vitamin K and prevent further bleeding in the coming days due to the long half-life of warfarin.

CLINICAL CORRELATION

Fresh Frozen Plasma (FFP) can also be used to reverse excessive bleeding caused by warfarin, however, it is less effective than PCC and it should only be used when PCC is not available.

Skin Necrosis. Warfarin can also cause **transient hypercoagulability** with increased clot format ove that warfarin inhibits the synthesis of the anticoagulant proteins C and S, which can cause a transient state of hypercoagulability. Patients with hereditary protein C or protein S deficiency are particularly susceptible to warfarin-induced skin necrosis because their anticoagulant protein levels are already low.

The hypercoagulability can cause a devastating adverse effect termed **warfarin-induced skin necrosis,** in which small vessels are thrombosed, causing skin to die (Figure 3). If skin necrosis occurs, it should be treated by stopping warfarin and giving vitamin K and PCC.

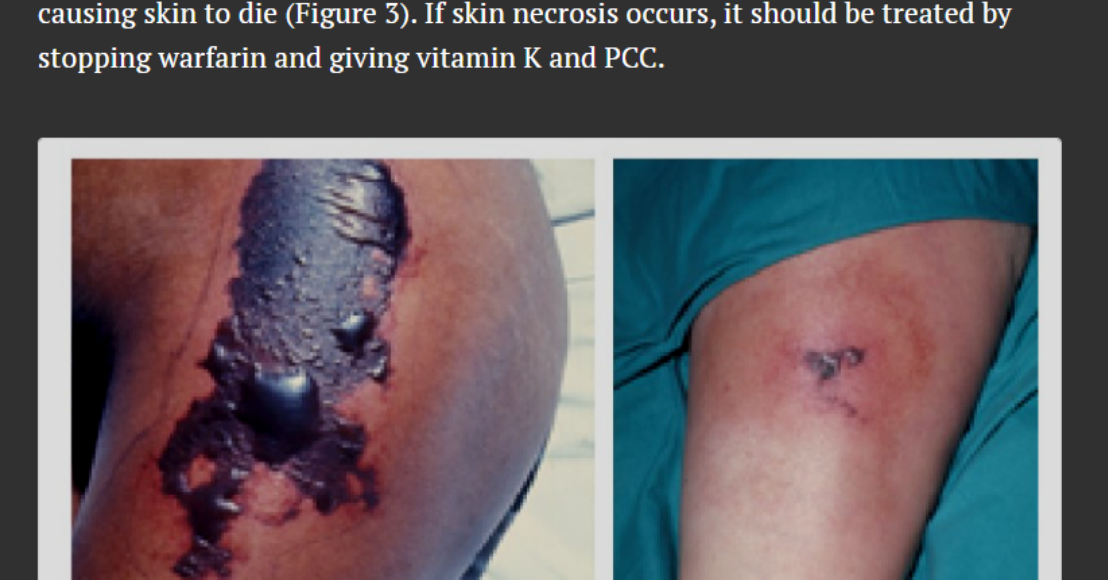

Figure 3

Contraindications

Warfarin is a *potent teratogen*, so this drug should be avoided during pregnancy. It also is contraindicated in patients with:

- Active bleeding or tendency to bleed
- Recent or potential surgery of the eye or CNS
- Unsupervised patients with conditions associated with a high potential for noncompliance

Drug-Drug Interactions

Warfarin is notorious for having many drug interactions. Not just with other medications, but with many foods too. Either way, the interactions will show on the patient's INR, hence the importance of monitoring.

Many foods are abundant in Vitamin K. These include green leafy vegetables (spinach, etc.), broccoli, kiwi, and many others. Patients taking warfarin need not avoid these foods, rather they need to be consistent on their intake of these foods. Too much Vitamin K intake could decrease the effectiveness of warfarin and cause a decreased or **subtherapeutic INR,** increasing the risk for clots.

Warfarin metabolism is primarily hepatic by **CYP 2C9**. As we have discussed in previous courses, this can result in numerous drug interactions. It is particularly concerning when a CYP 2C9 inhibitor is started which can result in a prolonged or **supratherapeutic INR,** increasing the risk for bleeding.

CLINICAL CORRELATION

A few common med aline, fluconazole, and amiodarone. All three of these drugs are inhibitors of CYP 2C9, and all of them can cause supratherapeutic INR and increase risk of bleeding.

Heparin

Heparin is a potent inhibitor of clot formation. It is commonly used for acute management of venous thromboembolism (VTE), as well as early in management of acute coronary syndrome. Heparin is also commonly given to patients in the hospital as prophylaxis, i.e., VTE prevention. There are two types of heparin:

- **High-molecular-weight heparin,** more commonly referred to as unfractionated heparin (UFH), which is given *intravenously* or *subcutaneously* (prophylaxis only). UFH activates antithrombin, inhibiting several clotting factors (II, IX, X, XI, and XII).
- **Low-molecular-weight heparin (LMWH),** including enoxaparin, dalteparin, and fondaparinux, which is given *subcutaneously*. LMWHs are smaller molecules than UFH and are active, fractionated fragments of the larger heparin molecule. LMWH is more selective than UFH, principally augmenting antithrombin inhibition of factor X.

Mechanism of Action

The various forms of heparin act **indirectly** on coagulation factors. They activate the endogenous anticoagulant antithrombin. **Antithrombin activation** then inhibits clotting factors Xa and IIa (thrombin), both in the common pathway (Figure 4).

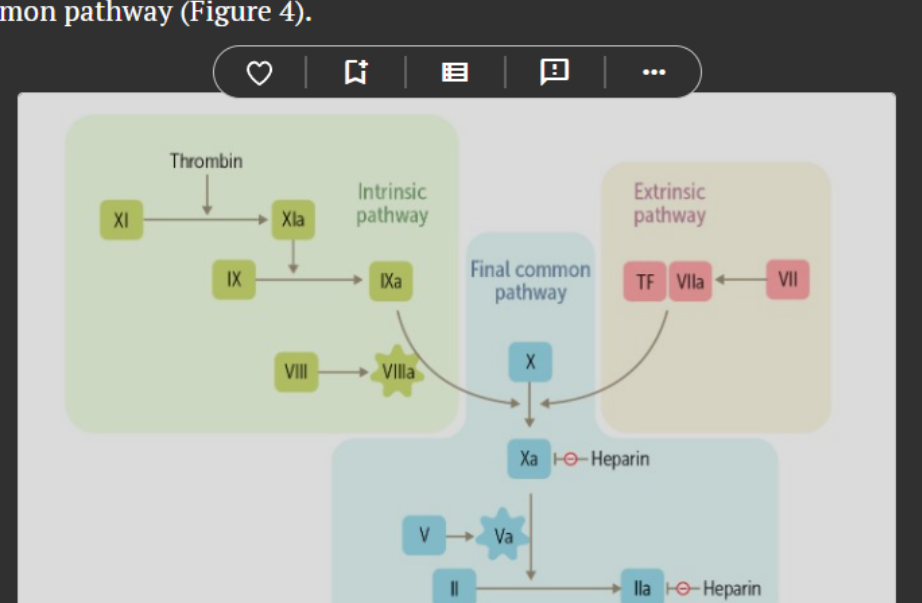

Figure 4

What is the mechanism of action of heparin anticoagulants?

Drug Monitoring and Dosing

As a drug that primarily inhibits the common pathway, heparin was/is traditionally monitored with activated PTT. However, many institutions have changed to monitoring a more specific heparin value, the anti-Xa level. This is because activated PTT can be more susceptible to inaccurate results due to other medications (aspirin, antihistamines, some antibiotics), whereas the anti-Xa level is more specific to the heparin activity. You will likely see both in your future practice.

Less-intensive monitoring is required for **LMWH.** Because of its subcutaneous administration, the absorption and anticoagulant effects are more predictable. Current monitoring (anti-Xa and/or aPTT) is recommended for high risk patients, or those of extreme body weights, and is becoming more routine at many institutions.

Adverse Effects

A common adverse effect of heparins is bleeding, just like with the other anticoagulants. Abnormal bleeding from heparin products can be treated with the antidote **protamine sulfate.** This is a positively charged molecule that binds to negatively charged heparin, inactivating it.

UFH also carries a risk of **heparin-induced thrombocytopenia and thrombosis (HITT).** This is an autoimmune reaction in which heparin binds to platelet factor 4 (PF4 n-PF4 complex; the triad of heparin-PF4-Ig antibody then activates platelets, causing clot formation (e.g., deep vein thrombosis) and thrombocytopenia. When HITT is suspected the patient needs to be switched immediately to a direct thrombin inhibitor (discussed below), which would be the anticoagulant of choice for a patient with HITT.

Contraindications

Contraindications to heparin include active bleeding, heparin allergy, or a history of HITT.

Direct Thrombin Inhibitors

Direct thrombin inhibitors include dabigatran, argatroban, and bivalirudin. Dabigatran is given orally while argatroban and bivalirudin are only given intravenously, and their primary indication is as alternative anticoagulants for patients with a history of HITT.

These agents inhibit thrombin directly (see Figure 5) and can act without a cofactor (e.g., antithrombin).

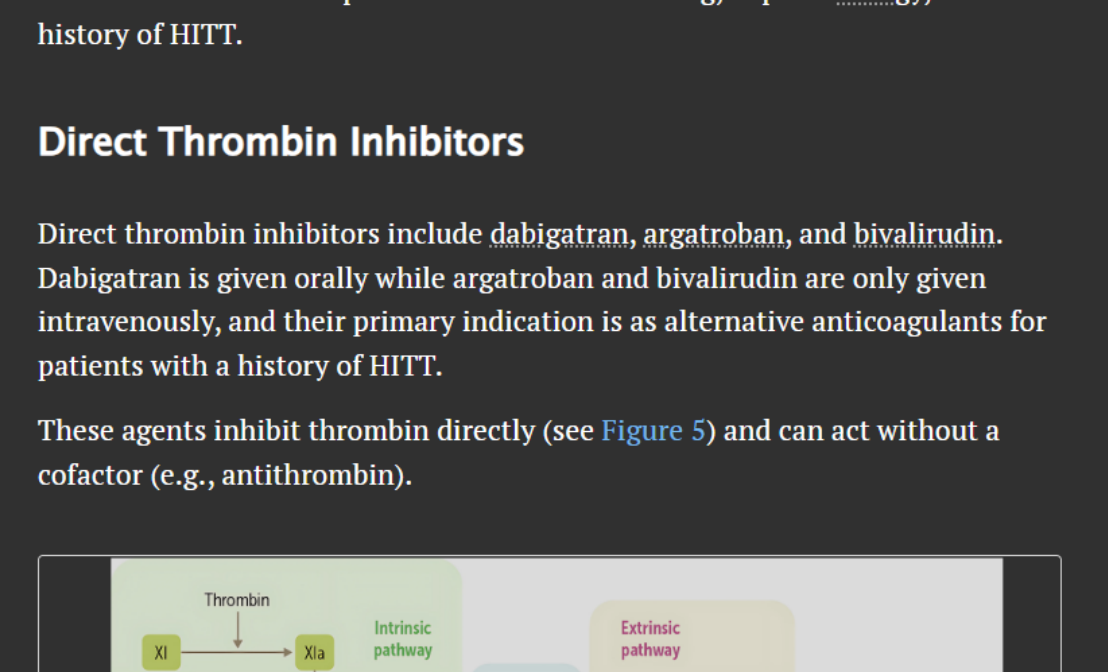

Figure 5

Adverse Effects

The most common adverse effect, as with the other anticoagulants is bleeding. Activated PCC is used in severe bleeding cases to reverse the effects of argatroban and bivalirudin.

Dabigatran, on the other hand, has a specific reversal agent called idarcizumab. This is a humanized monoclonal antibody fragment (Fab) that binds specifically to dabigatran and its acylglucuronide metabolites with an affinity for dabigatran that is ~350 times greater than that of thrombin. This neutralizes the anticoagulant effect within minutes.

Contraindications

Contraindications include hypersensitivity and active, heavy bleeding. Direct thrombin inhibitors should be used with caution in patients with liver disease because this is their primary route of metabolism.

CASE CONNECTION

BACK TO INTRODUCTION ↑

Thinking back to LK workup should be done?

You recognize three aspects, known as the Virchow triad, as the reason for LK's DVT: endothelial damage from surgical trauma, stasis from prolonged postoperative immobilization, and the hypercoagulable state of surgery. You explain each of these reasons to LK and how they led to the development of her blood clot. Because she is at high risk for DVT, and because this is her first one, an extensive workup is deemed unnecessary at this time. LK is treated with anticoagulants and responds well. She is discharged on postoperative day 6 and continues with outpatient physical therapy.

Summary

- Thrombotic disorders are conditions that increase the likelihood of abnormal blood clot formation, and can be either hereditary or acquired. Signs in a patient's history that would raise suspicion for a thrombotic disorder include personal history of recurrent thromboses, family history of thrombotic episodes, age <50 years, personal history of miscarriage, and presence of a venous thrombus somewhere other than the legs or lungs.
- Thrombi, or clots, can form within vessels and stay in place, or break off and move to other locations, becoming emboli.
- The Virchow triad, consisting of endothelial damage, abnormal blood flow, and hypercoagulability, summarizes the factors leading to thrombus formation.
- Common genetic thrombotic disorders include Factor V Leiden, Factor II gene mutation, antithrombin deficiency, and protein C and S deficiencies.
- Antiphospholipid thrombotic disorder where antibodies and excessive clotting.
- Anticoagulant therapy, including warfarin and heparin, is used to treat thrombotic disorders by inhibiting factors in the clotting cascade and requires careful monitoring due to potential adverse effects and interactions.
- Direct thrombin inhibitors, like argatroban and bivalirudin, are intravenous anticoagulants used for HITT, with bleeding as the most common side effect, and should be used cautiously in patients with liver disease, with activated PCC available for reversal in severe bleeding cases.

Review Questions

1. Thrombus formation in which of the following patients should most raise suspicion of a thrombotic disorder?

A. ○ 25-year-old pregnant patient

B. ● 42-year-old male construction worker

C. ○ Patient who takes oral contraceptives

D. ○ Patient with a thrombus in the deep veins of the leg

E. ○ Patient with a thrombus that occurs after major trauma

Hide Explanation

The correct answer is . Most thrombi occur in patients who have a combination of risk factors, such as smoking, oral contraceptive use, and/or immobilization. Although thrombotic disorders are rarely the culprit, certain scenarios should precipitate a search for an underlying thrombotic disorder, including young age (<50 years), recurrent thrombi, family history of thrombi, thrombi in unusual places, and a history of miscarriages. A thrombus in a 25-year-old pregnant patient (A) would not raise suspicion of an underlying disorder, given that pregnancy is a known risk factor for thrombi. Thrombus in a patient who takes oral contraceptives (C) would not raise suspicion of an underlying disorder, because taking oral contraceptives is a known risk factor for thrombi. The deep veins of the leg are very common places for thrombi, so a patient with a thrombus in the deep veins of the leg (D) would not raise suspicion of an underlying disorder. In contrast, thrombi in other places, such as the hepatic vein or the veins of the brain, make a thrombotic disorder more likely. A patient with a thrombus that occurs after major trauma (E) likely does not have an underlying disorder, because major trauma is a known risk factor for thrombi.

2. Which of the following thrombotic disorders is not hereditary?

A. ● Antiphospholipid antibody syndrome

B. ○ Antithrombin deficiency

C. ○ Factor V Leiden

D. ○ Protein C deficiency

E. ○ Protein S deficiency

Hide Explanation

The correct answer is antiphospholipid antibody syndrome (A). Antithrombin deficiency (B), factor V Leiden (C), protein C deficiency (D), and protein S deficiency (E) are all hereditary disorders.

3. A patient is given heparin to treat deep vein thrombosis and subsequently develops heavy gastrointestinal bleeding with hypotension. Which of the following would best reverse the action of heparin in this patient?

A. ○ Antithrombin

B. ○ Fresh frozen plasma

C. ● Protamine sulfate

D. ○ Tranexamic acid

E. ○ Vitamin K

Hide Explanation

The correct answer is protamine sulfate (C). Antithrombin (A) is an anticoagulant that is activated by heparin. Fresh frozen plasma (B) is an antidote to bleeds caused by warfarin (though prothrombin complex concentrates are preferred). Tranexamic acid (D) is an antidote for thrombolytic drugs. Vitamin K (E) is also an antidote to bleeds caused by wa

4. Which clotting factors are inhibited by heparin?

A. ○ IX and X

B. ○ VII and IX

C. ● Xa and IIa

D. ○ V and VIII

E. ○ I and XIII

Hide Explanation

The correct answer is Xa and IIa (C). Heparin activates the endogenous anticoagulant antithrombin. Antithrombin activation inhibits clotting factors Xa and IIa (C), which are part of the common pathway. Factors IX and X (A) are part of the intrinsic pathway. Factors VII and IX (B) are part of the extrinsic and intrinsic pathways, respectively. Factors V and VIII (D) are cofactors in the coagulation cascade, not directly inhibited by antithrombin. Factors I and XIII (E) are involved in the final steps of clot formation and stabilization.

How would you rate this Brick?

References

- Kumar V, Abbas AK, Aster JC, eds. *Robbins and Cotran Pathologic Basis of Disease.* 10th ed. Elsevier; 2021.

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Differentiate the clinical presentation of thrombotic disorders.~~

02 ☑ ~~List the types of thrombotic disorders and describe their basic underlying causes.~~

03 ☑ ~~Justify the therapeutic roles of warfarin, heparin, and direct thrombin inhibitors in treating thrombotic disorders.~~

04 ☑ ~~Recognize adverse drug effects, drug interactions, and contraindications of warfarin, heparin, and direct thrombin inhibitors.~~

Objective Confidence: 4/4

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